

Introduction to mathematics for economists

Exercise Sheet #1

Alexandre Grellet

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During the tutorials, we will solve exercises 1, 3, 5, 8, 10, and 12. So you must prioritize these. You will be provided a full answer sheet. It's normal to struggle with (at least) some exercises, don't panic, ask question and solve exercises with your classmates.

Continuity

Continuity and Uniform Continuity

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is **continuous at** x_0 if:

$$\forall \varepsilon > 0, \exists \delta > 0 : |x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon.$$

It is **uniformly continuous on a set** A if the same δ works for every point simultaneously:

$$\forall \varepsilon > 0, \exists \delta > 0 : \forall x, y \in A, |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

The key difference: in ordinary continuity δ may depend on both ε and x_0 ; in uniform continuity δ depends only on ε .

Exercise 1 : ε - δ proofs

1. Show that $f(x) = 3x + 2$ is continuous at every $x_0 \in \mathbb{R}$ using the ε - δ definition. (*Hint: just ask yourself what δ you must pick*)
2. Show that $g(x) = \frac{1}{x}$ is continuous at every $x_0 > 0$. *Note: this question is a bit hard, don't hesitate to collaborate.*
3. Show that $g(x) = \frac{1}{x}$ is **not** uniformly continuous on $(0, \infty)$. (*Hint: you may depart from the ε - δ style and find two sequences u_n and v_n such that $|u_n - v_n| \rightarrow 0$ but $|f(u_n) - f(v_n)|$ does not converge to 0.*)
4. Show that g is uniformly continuous on $[a, \infty)$ for any fixed $a > 0$. Explain why being bounded away from 0 matters in this case.

Solution

Question 1.

We want to show: given any $\varepsilon > 0$, we can find $\delta > 0$ such that $|x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon$.

Compute the difference directly:

$$|f(x) - f(x_0)| = |(3x + 2) - (3x_0 + 2)| = 3|x - x_0|.$$

So we need $3|x - x_0| < \varepsilon$, which holds whenever $|x - x_0| < \varepsilon/3$.

Choose $\delta = \varepsilon/3$. Then $|x - x_0| < \delta$ implies $|f(x) - f(x_0)| = 3|x - x_0| < 3 \cdot \frac{\varepsilon}{3} = \varepsilon$. ✓

Question 2.

Fix $x_0 > 0$. We need to find $\delta > 0$ depending on x_0 and ε such that $|x - x_0| < \delta \Rightarrow |g(x) - g(x_0)| < \varepsilon$.

Step 1: bound the difference. Compute:

$$|g(x) - g(x_0)| = \left| \frac{1}{x} - \frac{1}{x_0} \right| = \frac{|x - x_0|}{x \cdot x_0}.$$

To control this, we need to bound $\frac{1}{x}$ from above (we don't want the denominator to explode). So let's start by arbitrarily ruling out some possible δ s.

Restrict to $|x - x_0| < x_0/2$, which forces $3/2x_0 > x > x_0/2$, and in particular gives $x > x_0/2$. Then $\frac{1}{x} < \frac{2}{x_0}$, so:

$$|g(x) - g(x_0)| \leq \frac{2}{x_0^2} |x - x_0|.$$

Step 2: choose an actual δ . We need $\frac{2}{x_0^2} |x - x_0| < \varepsilon$, i.e. $|x - x_0| < \frac{\varepsilon x_0^2}{2}$. Set:

$$\delta = \min\left(\frac{x_0}{2}, \frac{\varepsilon x_0^2}{2}\right).$$

With this choice, both constraints hold simultaneously: $x > x_0/2$ and $|g(x) - g(x_0)| < \varepsilon$. ✓

Question 3.

To disprove uniform continuity, we exhibit two sequences with $|x_n - y_n| \rightarrow 0$ but $|g(x_n) - g(y_n)| \not\rightarrow 0$.

Take $x_n = \frac{1}{n}$ and $y_n = \frac{1}{n+1}$. Then:

$$|x_n - y_n| = \frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)} \rightarrow 0.$$

But:

$$|g(x_n) - g(y_n)| = |n - (n+1)| = 1 \not\rightarrow 0.$$

So taking $\varepsilon_0 = 1/2$, we have found sequences with $|x_n - y_n| \rightarrow 0$ yet $|g(x_n) - g(y_n)| \geq \varepsilon_0$ for every n . No single δ can work for all pairs simultaneously: near zero, g is so steep that any fixed δ is too coarse. ✓

Question 4.

On $[a, \infty)$ with $a > 0$, for any $x, y \geq a$:

$$|g(x) - g(y)| = \frac{|x - y|}{xy} \leq \frac{|x - y|}{a^2}.$$

The key step: since $x, y \geq a > 0$, the product $xy \geq a^2$ is bounded *away from zero uniformly* across the entire domain. Choose $\delta = a^2\varepsilon$, which depends only on ε and a , not on the particular x . Then $|x - y| < \delta$ implies $|g(x) - g(y)| < \varepsilon$. ✓

On $(0, \infty)$, no such lower bound on xy exists: as $x, y \rightarrow 0^+$, the denominator xy can be made arbitrarily small, which is exactly what the sequences in Question 3 exploit. The set $[a, \infty)$ is not compact (it is not bounded), but it is closed and bounded away from zero, which is what matters here.

Exercise 2 : Sequential continuity and a demand function

In economics, a **demand function** $D(p)$ maps a price $p > 0$ to the quantity of a good that consumers wish to purchase at that price. We consider:

$$D(p) = \frac{A}{p^\eta}, \quad p > 0,$$

where $A > 0$ is a scale parameter and $\eta > 0$ is the **price elasticity of demand** (a higher η means consumers are more sensitive to price changes).

1. Using the **sequential definition** of continuity, show that D is continuous on $(0, \infty)$: if $p_n \rightarrow p$ with $p > 0$, then $D(p_n) \rightarrow D(p)$. You may need to use the following theorem:

Theorem

If $f : K \rightarrow V$ is continuous and $g : V \rightarrow S$ is continuous, then $g \circ f$ is continuous.

2. Let $\eta = 1$, $A = 1$. Construct an explicit sequence $p_n \rightarrow 0^+$ and show $D(p_n) \rightarrow +\infty$. What does this say about market behavior when prices approach zero?
3. Is D uniformly continuous on $(0, \infty)$? On $[\underline{p}, \bar{p}]$ for fixed $0 < \underline{p} < \bar{p}$? Justify briefly.

Solution

Question 1.

Suppose $p_n \rightarrow p$ with $p > 0$. We apply the algebra of limits step by step:

- Since $p_n \rightarrow p$ and $\eta > 0$, we have $p_n^\eta \rightarrow p^\eta$ (power functions are continuous).
- Since $p^\eta > 0$, it follows that $\frac{1}{p_n^\eta} \rightarrow \frac{1}{p^\eta}$ (the reciprocal function is continuous away from zero).
- Multiplying by the constant $A > 0$: $D(p_n) = \frac{A}{p_n^\eta} \rightarrow \frac{A}{p^\eta} = D(p)$.

This confirms sequential continuity. ✓

Question 2.

Take $p_n = 1/n$, so $p_n \rightarrow 0^+$. With $\eta = 1$ and $A = 1$:

$$D(p_n) = \frac{1}{1/n} = n \rightarrow +\infty.$$

Economic interpretation. As the price of a good approaches zero, consumers wish to buy infinite quantities. This is not physically feasible and signals that this demand specification is only meaningful for prices bounded away from zero. In practice, markets always operate with $p \geq \underline{p} > 0$ for some minimum price.

Question 3.

On $(0, \infty)$: not uniformly continuous. The derivative is $D'(p) = -A\eta p^{-\eta-1}$, so $|D'(p)| \rightarrow +\infty$ as $p \rightarrow 0^+$. The function becomes arbitrarily steep near zero, meaning that for any fixed $\delta > 0$, pairs (p, q) with $|p - q| < \delta$ near zero can have $|D(p) - D(q)|$ arbitrarily large.
On $[\underline{p}, \bar{p}]$: yes, uniformly continuous. The interval is closed and bounded, hence compact. By the Heine-Cantor theorem, any function continuous on a compact set is automatically uniformly continuous. No further argument is needed.

Sequences

Convergence of a Sequence

A sequence $(a_n)_{n \geq 1}$ **converges to** $L \in \mathbb{R}$, written $a_n \rightarrow L$, if:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} : n > N \implies |a_n - L| < \varepsilon.$$

Think of it as a game: your opponent picks any tolerance $\varepsilon > 0$, however tiny, and you must produce a threshold N after which every term of the sequence lies within ε of L .

Exercise 3 : Convergence from the definition

Let $a_n = \frac{3n+1}{n+2}$.

1. Compute the first five terms of the sequence and conjecture the limit L .
2. Prove your conjecture rigorously using the ε - N definition. (*Hint: simplify $|a_n - L|$ algebraically first.*)
3. Find explicitly the smallest $N(\varepsilon)$ such that $n > N(\varepsilon)$ guarantees $|a_n - L| < \varepsilon$.

Solution

Question 1.

$$a_1 = \frac{4}{3} \approx 1.33, \quad a_2 = \frac{7}{4} = 1.75, \quad a_3 = \frac{10}{5} = 2, \quad a_4 = \frac{13}{6} \approx 2.17, \quad a_5 = \frac{16}{7} \approx 2.29.$$

The terms are increasing and appear to approach 3. Conjecture: $L = 3$.

Question 2.

We need to show: for any $\varepsilon > 0$, there exists N such that $n > N \implies |a_n - 3| < \varepsilon$.

Step 1: simplify $|a_n - L|$.

$$|a_n - 3| = \left| \frac{3n+1}{n+2} - 3 \right| = \left| \frac{3n+1-3(n+2)}{n+2} \right| = \left| \frac{3n+1-3n-6}{n+2} \right| = \frac{5}{n+2}.$$

Step 2: find when this is less than ε .

We need $\frac{5}{n+2} < \varepsilon$, which rearranges to:

$$n > \frac{5}{\varepsilon} - 2.$$

Step 3: choose N .

Set $N = \left\lceil \frac{5}{\varepsilon} - 2 \right\rceil$. Then for all $n > N$:

$$|a_n - 3| = \frac{5}{n+2} < \varepsilon. \quad \checkmark$$

Question 3.

From the calculation above: $N(\varepsilon) = \left\lceil \frac{5}{\varepsilon} - 2 \right\rceil$.

For instance, to guarantee $|a_n - 3| < 0.01$, we need $n > 5/0.01 - 2 = 498$, so $N = 498$.

Exercise 4 : Monotone convergence and the harmonic series (difficult)

Monotone Convergence Theorem

A sequence that is **increasing** ($a_{n+1} \geq a_n$) and **bounded above** converges. A sequence that is **decreasing** ($a_{n+1} \leq a_n$) and **bounded below** converges. You do *not* need to know the limit in advance.

Consider the two sequences of partial sums:

$$S_n = \sum_{k=1}^n \frac{1}{k^2}, \quad H_n = \sum_{k=1}^n \frac{1}{k}.$$

1. Show that both (S_n) and (H_n) are strictly increasing.
2. Show that (S_n) is bounded above. (*Hint: show $\frac{1}{k^2} \leq \frac{1}{k(k-1)} = \frac{1}{k-1} - \frac{1}{k}$ for $k \geq 2$, then telescope.*)
3. Conclude by the Monotone Convergence Theorem that (S_n) converges. (Its limit is $\pi^2/6$, a famous result you may use without proof.)
4. Show that (H_n) is **not** bounded above, and hence diverges. (*Hint: group terms as $H_{2n} \geq 1 + n/2$, by noting that each doubling block contributes at least $1/2$.*)
5. What does the divergence of (H_n) tell you about the statement “each individual term $1/k$ goes to zero”? Is this sufficient for a series to converge?

Solution

Question 1.

Both sequences increase because each new term is strictly positive:

$$S_{n+1} - S_n = \frac{1}{(n+1)^2} > 0, \quad H_{n+1} - H_n = \frac{1}{n+1} > 0.$$

So both are strictly increasing. ✓

Question 2.

For $k \geq 2$, observe:

$$\frac{1}{k^2} \leq \frac{1}{k(k-1)} = \frac{1}{k-1} - \frac{1}{k}.$$

The second equality is partial fractions. Summing from $k = 2$ to n :

$$\sum_{k=2}^n \frac{1}{k^2} \leq \sum_{k=2}^n \left(\frac{1}{k-1} - \frac{1}{k} \right).$$

The right-hand side is a telescoping sum:

$$\sum_{k=2}^n \left(\frac{1}{k-1} - \frac{1}{k} \right) = \frac{1}{1} - \frac{1}{n} = 1 - \frac{1}{n} < 1.$$

Therefore:

$$S_n = 1 + \sum_{k=2}^n \frac{1}{k^2} < 1 + 1 = 2.$$

So $S_n < 2$ for all n . ✓

Question 3.

(S_n) is strictly increasing (Question 1) and bounded above by 2 (Question 2). By the Monotone Convergence Theorem, it converges. Its limit is the famous Basel sum $\pi^2/6 \approx 1.645$. ✓

Question 4.

We group the terms of H_n in blocks whose size doubles at each step. Start from $H_1 = 1$, then consider the subsequence H_{2^n} for $n \geq 1$.

Write:

$$H_{2^n} = 1 + \frac{1}{2} + \underbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}_{\geq 2 \cdot \frac{1}{4} = \frac{1}{2}} + \underbrace{\left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right)}_{\geq 4 \cdot \frac{1}{8} = \frac{1}{2}} + \cdots + \underbrace{\left(\frac{1}{2^{n-1}+1} + \cdots + \frac{1}{2^n}\right)}_{\geq 2^{n-1} \cdot \frac{1}{2^n} = \frac{1}{2}}.$$

The structure is: after the initial term 1, there are exactly n blocks — the k -th block consists of the 2^{k-1} terms from $\frac{1}{2^{k-1}+1}$ to $\frac{1}{2^k}$, and each of these terms is at least $\frac{1}{2^k}$, so the block sum is at least $2^{k-1} \cdot \frac{1}{2^k} = \frac{1}{2}$. Therefore:

$$H_{2^n} \geq 1 + \underbrace{\frac{1}{2} + \frac{1}{2} + \cdots + \frac{1}{2}}_{n \text{ blocks}} = 1 + \frac{n}{2} \rightarrow +\infty.$$

Since the subsequence (H_{2^n}) diverges to $+\infty$, the full sequence (H_n) is unbounded and therefore diverges. ✓

Question 5.

The terms $1/k \rightarrow 0$, yet $\sum_{k=1}^{\infty} 1/k = +\infty$. This shows that having individual terms go to zero is **necessary** but **not sufficient** for a series to converge. The harmonic series is the canonical counterexample, and it is worth remembering.

Exercise 5 : A price adjustment process

Fixed Point

A **fixed point** of a function h is a value x^* such that $h(x^*) = x^*$. In economics, fixed points often represent equilibria: states that are self-sustaining once reached.

A market for a commodity adjusts prices over time according to the rule:

$$p_{n+1} = \frac{1}{3}p_n + 8, \quad p_0 = 0.$$

Think of p_n as the market price in period n : each period, the price partially reverts toward a long-run value of 12.

1. Compute p_1, p_2, p_3, p_4, p_5 . Is the sequence increasing? (prove or disprove)
2. Show the sequence is bounded above by 12. (*Hint: show by induction that $p_n \leq 12$ for all n .*)
3. Conclude using the Monotone Convergence Theorem that (p_n) converges to some limit p^* .
4. Find p^* by taking limits on both sides of the recursion. Interpret p^* as the long-run equilibrium price.

i Solution

Question 1.

$$p_0 = 0, \quad p_1 = 8, \quad p_2 = \frac{32}{3} \approx 10.67, \quad p_3 = \frac{104}{9} \approx 11.56, \quad p_4 = \frac{320}{27} \approx 11.85, \quad p_5 = \frac{968}{81} \approx 11.95.$$

The sequence is strictly increasing: each term is closer to 12 than the last.

Question 2.

We prove $p_n \leq 12$ for all n by induction.

Base case. $p_0 = 0 \leq 12$. ✓

Inductive step. Assume $p_n \leq 12$. Then:

$$p_{n+1} = \frac{1}{3}p_n + 8 \leq \frac{1}{3} \cdot 12 + 8 = 4 + 8 = 12. \quad \checkmark$$

Question 3.

We check the two conditions of the Monotone Convergence Theorem:

Monotone. $p_{n+1} - p_n = \frac{1}{3}p_n + 8 - p_n = 8 - \frac{2}{3}p_n = \frac{2}{3}(12 - p_n) \geq 0$ since $p_n \leq 12$. So the sequence is non-decreasing.

Bounded above. By Question 2, $p_n \leq 12$ for all n .

By the Monotone Convergence Theorem, (p_n) converges to some limit $p^* \leq 12$. ✓

Question 4.

Taking limits on both sides of $p_{n+1} = \frac{1}{3}p_n + 8$ as $n \rightarrow \infty$:

$$p^* = \frac{1}{3}p^* + 8 \implies \frac{2}{3}p^* = 8 \implies p^* = 12.$$

Economic interpretation. The price $p^* = 12$ is the unique fixed point of the adjustment rule: once reached, the market stays there. The process models how a market corrects toward equilibrium step by step. The coefficient $1/3$ determines how quickly the correction happens: the smaller it is, the faster prices converge.

Rates of Convergence

Big- O and Little- o Notation

Let (a_n) and (b_n) be sequences with $b_n > 0$.

- $a_n = O(b_n)$: there exists $C > 0$ and N such that $n > N \Rightarrow |a_n| \leq Cb_n$. We say a_n is *at most of order* b_n .
- $a_n = o(b_n)$: $a_n/b_n \rightarrow 0$. We say a_n is *negligible relative to* b_n .

Key relationship: $o(b_n) \Rightarrow O(b_n)$, but not conversely.

Exercise 6 : Classifying rates

Classify each sequence below as $O(n^{-1})$, $o(n^{-1})$, $O(n^{-1/2})$, $o(n^{-1/2})$, or none of these, as $n \rightarrow \infty$. A sequence may belong to several categories. Justify every answer.

$$a_n = \frac{2}{n}, \quad b_n = \frac{\ln n}{n}, \quad c_n = \frac{1}{n^{3/4}}, \quad d_n = \frac{1}{\sqrt{n} \ln n}, \quad e_n = \frac{1}{n^2}$$

i Solution

The strategy for each sequence is to compute the ratio with the benchmark and take the limit. Recall: if the ratio converges to a finite nonzero constant, the sequence is O but not o ; if the ratio goes to 0, it is o (and hence also O); if the ratio goes to ∞ , it is neither.

Summary table:

Sequence	$O(n^{-1/2})$	$o(n^{-1/2})$	$O(n^{-1})$	$o(n^{-1})$
$2/n$	Yes	Yes	Yes	No
$(\ln n)/n$	Yes	Yes	No	No
$n^{-3/4}$	Yes	No	No	No
$1/(\sqrt{n} \ln n)$	Yes	Yes	No	No
n^{-2}	Yes	Yes	Yes	Yes

Justifications:

$$a_n = 2/n.$$

$$\frac{a_n}{n^{-1}} = 2 \rightarrow 2 \neq 0 \Rightarrow O(n^{-1}) \text{ but not } o(n^{-1}).$$

$$\frac{a_n}{n^{-1/2}} = \frac{2}{\sqrt{n}} \rightarrow 0 \Rightarrow o(n^{-1/2}).$$

$$b_n = (\ln n)/n.$$

$$\frac{b_n}{n^{-1}} = \ln n \rightarrow \infty \Rightarrow \text{not } O(n^{-1}) \text{ (grows faster than } 1/n).$$

$$\frac{b_n}{n^{-1/2}} = \frac{\ln n}{\sqrt{n}} \rightarrow 0 \Rightarrow o(n^{-1/2}).$$

$$c_n = n^{-3/4}.$$

$$\frac{c_n}{n^{-1}} = n^{1/4} \rightarrow \infty \Rightarrow \text{not } O(n^{-1}).$$

$$\frac{c_n}{n^{-1/2}} = n^{-1/4} \rightarrow 0 \Rightarrow o(n^{-1/2}).$$

Note: $c_n = n^{-3/4}$ is $o(n^{-1/2})$ and hence also $O(n^{-1/2})$. The table entry above has been corrected from the draft: c_n is *not* merely $O(n^{-1/2})$ — it is $o(n^{-1/2})$ as well, since the ratio $n^{-1/4} \rightarrow 0$.

$$d_n = 1/(\sqrt{n} \ln n).$$

$$\frac{d_n}{n^{-1/2}} = \frac{1}{\ln n} \rightarrow 0 \Rightarrow o(n^{-1/2}).$$

$$\frac{d_n}{n^{-1}} = \frac{\sqrt{n}}{\ln n} \rightarrow \infty \Rightarrow \text{not } O(n^{-1}).$$

$$e_n = n^{-2}.$$

$$\frac{e_n}{n^{-1}} = n^{-1} \rightarrow 0 \Rightarrow o(n^{-1}) \text{ (and hence also } O(n^{-1})).$$

$$\frac{e_n}{n^{-1/2}} = n^{-3/2} \rightarrow 0 \Rightarrow o(n^{-1/2}).$$

Exercise 7 : Taylor remainder and approximation quality

Consider the function $f(x) = \sqrt{1+x}$, expanded around $x = 0$.

1. Compute the first-order Taylor expansion $P_1(x)$ of f around $x = 0$.
2. Show that the remainder $R_1(x) = f(x) - P_1(x)$ satisfies $R_1(x) = o(x)$ as $x \rightarrow 0$. (*This is exactly what Taylor's theorem promises.*)
3. Show that $R_1(x) = O(x^2)$ as $x \rightarrow 0$ but $R_1(x) \neq o(x^2)$. (*Compute $R_1(x)/x^2$ and take the limit.*)
4. A firm's marginal cost is approximated as $MC(q) \approx 1 + \frac{q}{2}$ for small q , where the true marginal cost is $MC(q) = \sqrt{1+q}$. Suppose $q = O(n^{-1/2})$ (the quantity shrinks as the market grows). What is the order of the approximation error $MC(q) - (1 + q/2)$ in terms of n ? Do you think this is good enough for an actual firm?

i Solution

Question 1.

Compute the derivatives at $x = 0$:

$$f(x) = (1+x)^{1/2}, \quad f(0) = 1.$$

$$f'(x) = \frac{1}{2}(1+x)^{-1/2}, \quad f'(0) = \frac{1}{2}.$$

The first-order Taylor polynomial is:

$$P_1(x) = f(0) + f'(0)x = 1 + \frac{x}{2}.$$

Question 2.

The remainder is $R_1(x) = \sqrt{1+x} - 1 - \frac{x}{2}$. We check whether $R_1(x)/x \rightarrow 0$:

$$\frac{R_1(x)}{x} = \frac{\sqrt{1+x} - 1}{x} - \frac{1}{2}.$$

The first term is a difference quotient. As $x \rightarrow 0$:

$$\frac{\sqrt{1+x} - 1}{x} = \frac{f(x) - f(0)}{x - 0} \rightarrow f'(0) = \frac{1}{2}.$$

Therefore:

$$\frac{R_1(x)}{x} \rightarrow \frac{1}{2} - \frac{1}{2} = 0,$$

which confirms $R_1(x) = o(x)$. ✓

Question 3.

Compute $f''(x) = -\frac{1}{4}(1+x)^{-3/2}$, so $f''(0) = -\frac{1}{4}$.

By the second-order Taylor theorem:

$$R_1(x) = f(x) - P_1(x) = \frac{f''(0)}{2}x^2 + o(x^2) = -\frac{x^2}{8} + o(x^2).$$

Therefore:

$$\frac{R_1(x)}{x^2} \rightarrow -\frac{1}{8} \neq 0.$$

This shows $R_1(x) = O(x^2)$ (the ratio is bounded) but $R_1(x) \neq o(x^2)$ (the ratio does not go to zero). The remainder is exactly of order x^2 .

Question 4.

From Question 3, the approximation error satisfies $R_1(q) = O(q^2)$ as $q \rightarrow 0$.

Since $q = O(n^{-1/2})$, we have $q^2 = O(n^{-1})$. Therefore:

$$MC(q) - \left(1 + \frac{q}{2}\right) = R_1(q) = O(q^2) = O(n^{-1}).$$

The approximation error shrinks at rate n^{-1} , which is faster than q itself (which shrinks at $n^{-1/2}$). This confirms that the linear approximation is accurate in relative terms: the error is of smaller order than the quantity being approximated.

Linearization (Univariate)

Exercise 8 : Bond pricing, duration, and convexity

In financial economics, a **zero-coupon bond** is a contract that pays exactly \$1 at a future date T (called the **maturity**). If the continuously compounded interest rate is $r > 0$, the price of this bond today is:

$$P(r) = e^{-rT}.$$

A lower interest rate means money in the future is more valuable today, so P decreases in r .

1. Compute $P'(r)$ and $P''(r)$. Verify that P is decreasing and convex in r .
2. Write the second-order Taylor expansion of $P(r)$ around a given rate r_0 :

$$P(r) \approx P(r_0) + P'(r_0)(r - r_0) + \frac{1}{2}P''(r_0)(r - r_0)^2.$$

Show that this can be written as:

$$P(r) \approx P(r_0) \left[1 - T(r - r_0) + \frac{T^2}{2}(r - r_0)^2 \right].$$

The coefficient T is called the **duration** of the bond.

3. Interpret the linear term: if r rises by 1 percentage point (i.e. $r - r_0 = 0.01$), by approximately how much does the bond price fall, in percentage terms? Why does a longer-maturity bond fall more?
4. The second-order term captures **convexity**: the price curve bends upward. Show that the true price $P(r)$ is always **above** the linear approximation for r around r_0 . You will need to show that $e^u > 1 + u$ for all $u \neq 0$. (*Hint: MVT could prove useful*)
5. What does convexity imply for an investor who holds bonds and faces uncertain interest rates? Is convexity good or bad for the investor? Explain in one or two sentences.

Solution

Question 1.

Differentiating $P(r) = e^{-rT}$:

$$P'(r) = -Te^{-rT} < 0 \quad (\text{decreasing in } r).$$

$$P''(r) = T^2 e^{-rT} > 0 \quad (\text{convex in } r).$$

Both signs follow because $T > 0$ and $e^{-rT} > 0$ for all r . ✓

Question 2.

Evaluate the derivatives at r_0 :

$$P(r_0) = e^{-r_0T}, \quad P'(r_0) = -Te^{-r_0T}, \quad P''(r_0) = T^2 e^{-r_0T}.$$

Substituting into the Taylor formula:

$$P(r) \approx e^{-r_0T} - Te^{-r_0T}(r - r_0) + \frac{T^2}{2} e^{-r_0T}(r - r_0)^2.$$

Factoring out $e^{-r_0T} = P(r_0)$:

$$P(r) \approx P(r_0) \left[1 - T(r - r_0) + \frac{T^2}{2}(r - r_0)^2 \right]. \quad \checkmark$$

Question 3.

From the linear term, a small change $\Delta r = r - r_0$ causes:

$$\frac{\Delta P}{P(r_0)} \approx -T \cdot \Delta r.$$

For $\Delta r = 0.01$ (a one percentage point rise):

$$\frac{\Delta P}{P(r_0)} \approx -T \times 0.01.$$

A bond with maturity $T = 10$ years loses about 10% of its value; one with $T = 2$ years loses about 2%. The intuition is that a longer bond has more cash flows far in the future, and those are discounted more heavily when rates rise.

Question 4.

We prove $e^u > 1 + u$ for all $u \neq 0$ using the Mean Value Theorem in both cases.

Case $u > 0$. Apply the MVT to $f(x) = e^x$ on $[0, u]$. There exists $c \in (0, u)$ such that:

$$e^u - e^0 = e^c \cdot (u - 0) \implies e^u = 1 + u e^c.$$

Since $c > 0$, we have $e^c > e^0 = 1$, so $e^u = 1 + u e^c > 1 + u$. ✓

Case $u < 0$. Apply the MVT to $f(x) = e^x$ on $[u, 0]$. There exists $c \in (u, 0)$ such that:

$$e^0 - e^u = e^c \cdot (0 - u) \implies 1 - e^u = -u e^c.$$

Since $c < 0$, we have $e^c < e^0 = 1$, so $-u e^c < -u \cdot 1 = -u$ (recall $-u > 0$), which gives $1 - e^u < -u$, i.e. $e^u > 1 + u$. ✓

Setting $u = -T(r - r_0)$, the claim $P(r) > L(r)$ (where L is the linear approximation) follows immediately.

Question 5.

Convexity is **good** for the bond investor. When rates fall, the bond price rises by more than the linear approximation predicts; when rates rise, the price falls by less. The investor benefits from an asymmetry: gains exceed losses for equal-sized rate movements in either direction.

Multivariate Calculus

Gradient and Jacobian

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be C^1 . The **gradient** of f at x^* is the vector of partial derivatives:

$$\nabla f(x^*) = \left(\frac{\partial f}{\partial x_1}(x^*), \dots, \frac{\partial f}{\partial x_n}(x^*) \right)^\top.$$

For $F = (f_1, \dots, f_m) : \mathbb{R}^n \rightarrow \mathbb{R}^m$, the **Jacobian** is the $m \times n$ matrix:

$$J_F(x^*) = \begin{pmatrix} \partial f_1 / \partial x_1 & \cdots & \partial f_1 / \partial x_n \\ \vdots & \ddots & \vdots \\ \partial f_m / \partial x_1 & \cdots & \partial f_m / \partial x_n \end{pmatrix}_{x=x^*}.$$

Exercise 9 : Gradient and isoprofit curves (difficult)

A firm's profit is:

$$\pi(K, L) = K^{1/3} L^{1/3} - wL - rK,$$

where $K > 0$ is capital, $L > 0$ is labor, $w > 0$ is the **wage** (cost per unit of labor), and $r > 0$ is the **rental rate of capital** (cost per unit of capital). These are fixed parameters, not variables.

1. Compute the gradient $\nabla \pi(K, L)$ at a general point (K, L) .
2. Evaluate $\nabla \pi$ at the specific point $(K, L) = (1, 1)$ when $w = r = \frac{1}{3}$.
3. An **isoprofit curve** is the set of all (K, L) pairs that give the same profit level $\bar{\pi}$:

$$\mathcal{J} = \{(K, L) \in \mathbb{R}_{++}^2 : \pi(K, L) = \bar{\pi}\}.$$

Without doing any computation, explain why $\nabla \pi$ is said to be **orthogonal** to $\mathcal{J}$ at any point on it. (*Hint: think about what it means to move along $\mathcal{J}$.*)

4. At the point $(1, 1)$ with $w = r = 1/3$, you found $\nabla \pi(1, 1) = (0, 0)^\top$. This means $(1, 1)$ is a **critical point** of π : the gradient vanishes, and the standard orthogonality argument breaks down (the zero vector is orthogonal to everything, which gives no useful information about the geometry). To see the gradient-isoprofit orthogonality, move to a nearby point where the gradient is nonzero. Take $(K, L) = (2, 1)$ with the same $w = r = 1/3$.
 - (a) Compute $\nabla \pi(2, 1)$.
 - (b) The isoprofit curve through $(2, 1)$ is implicitly defined by $\pi(K, L) = \pi(2, 1)$. Differentiate this constraint implicitly with respect to K to find $\frac{dL}{dK}$ at $(2, 1)$, i.e. the slope of the tangent to the isoprofit curve.
 - (c) Write down the tangent direction vector $v = (1, dL/dK)$ and verify by direct computation that $\nabla \pi(2, 1) \cdot v = 0$.

Solution

Question 1.

Differentiate π with respect to each input:

$$\frac{\partial \pi}{\partial K} = \frac{1}{3} K^{-2/3} L^{1/3} - r, \quad \frac{\partial \pi}{\partial L} = \frac{1}{3} K^{1/3} L^{-2/3} - w.$$

So the gradient is:

$$\nabla\pi(K, L) = \begin{pmatrix} \frac{L^{1/3}}{3K^{2/3}} - r \\ \frac{K^{1/3}}{3L^{2/3}} - w \end{pmatrix}.$$

Each partial has a clean interpretation: $\frac{\partial\pi}{\partial K}$ is the marginal product of capital minus its rental cost; $\frac{\partial\pi}{\partial L}$ is the marginal product of labor minus the wage.

Question 2.

At $(K, L) = (1, 1)$ with $w = r = 1/3$:

$$\left. \frac{\partial\pi}{\partial K} \right|_{(1,1)} = \frac{1}{3} - \frac{1}{3} = 0, \quad \left. \frac{\partial\pi}{\partial L} \right|_{(1,1)} = \frac{1}{3} - \frac{1}{3} = 0.$$

So $\nabla\pi(1, 1) = (0, 0)^\top$. This is a **critical point** of π : the marginal product of each input exactly equals its cost, so the firm has no incentive to adjust either input at the margin. We will study in a later session what kind of critical point this is (maximum, minimum, or saddle point).

Question 3.

Moving along $\mathcal{I}$ means following a path $(K(t), L(t))$ that keeps profit constant: $\pi(K(t), L(t)) = \bar{\pi}$ for all t . Differentiating with respect to t :

$$\frac{d}{dt}\pi(K(t), L(t)) = \nabla\pi \cdot \begin{pmatrix} \dot{K} \\ \dot{L} \end{pmatrix} = 0.$$

The vector $(\dot{K}, \dot{L})$ is the tangent to the curve at that point. So $\nabla\pi \cdot v = 0$ for every tangent direction v , which is exactly the statement that $\nabla\pi$ is orthogonal to $\mathcal{I}$.

Economic intuition. The gradient points in the direction of steepest profit increase. Moving along an isoprofit curve neither increases nor decreases profit, so we move perpendicular to the gradient.

Question 4.

(a) At $(K, L) = (2, 1)$ with $w = r = 1/3$:

$$\left. \frac{\partial\pi}{\partial K} \right|_{(2,1)} = \frac{1^{1/3}}{3 \cdot 2^{2/3}} - \frac{1}{3} = \frac{1}{3 \cdot 2^{2/3}} - \frac{1}{3} = \frac{1}{3} \left(\frac{1}{2^{2/3}} - 1 \right).$$

$$\left. \frac{\partial\pi}{\partial L} \right|_{(2,1)} = \frac{2^{1/3}}{3 \cdot 1^{2/3}} - \frac{1}{3} = \frac{2^{1/3}}{3} - \frac{1}{3} = \frac{1}{3} (2^{1/3} - 1).$$

So:

$$\nabla\pi(2, 1) = \frac{1}{3} \begin{pmatrix} 2^{-2/3} - 1 \\ 2^{1/3} - 1 \end{pmatrix}.$$

Note $2^{-2/3} - 1 < 0$ and $2^{1/3} - 1 > 0$, so the gradient is nonzero. ✓

(b) Differentiate $\pi(K, L(K)) = \bar{\pi}$ implicitly with respect to K :

$$\frac{\partial\pi}{\partial K} + \frac{\partial\pi}{\partial L} \cdot \frac{dL}{dK} = 0 \quad \implies \quad \frac{dL}{dK} = -\frac{\partial\pi/\partial K}{\partial\pi/\partial L}.$$

At $(2, 1)$:

$$\left. \frac{dL}{dK} \right|_{(2,1)} = -\frac{2^{-2/3} - 1}{2^{1/3} - 1}.$$

(c) The tangent direction is $v = \left(1, -\frac{2^{-2/3} - 1}{2^{1/3} - 1}\right)$. Compute the dot product:

$$\begin{aligned}\nabla\pi(2, 1) \cdot v &= \frac{1}{3} \left[(2^{-2/3} - 1) \cdot 1 + (2^{1/3} - 1) \cdot \left(-\frac{2^{-2/3} - 1}{2^{1/3} - 1}\right) \right] \\ &= \frac{1}{3} [(2^{-2/3} - 1) - (2^{-2/3} - 1)] = 0. \quad \checkmark\end{aligned}$$

The gradient is indeed orthogonal to the isoprofit curve at $(2, 1)$.

Exercise 10 : Jacobian of an excess demand system

When consumers want to buy more of a good than is available, we say there is **excess demand**; when there is more available than consumers want, there is **excess supply**. A market is in **equilibrium** when neither occurs: the quantity consumers wish to buy exactly equals the quantity supplied. We call this an equilibrium price because, once reached, there is no pressure for the price to change: every seller finds a buyer at that price, and every buyer finds a seller.

Consider the following two-good excess demand system:

$$Z_1(p_1, p_2) = \frac{p_2}{p_1} - 1, \quad Z_2(p_1, p_2) = \frac{p_1}{p_2} - 1.$$

Here $Z_i > 0$ means consumers want more of good i than is available (excess demand); $Z_i < 0$ means there is a surplus.

1. Verify that $(p_1, p_2) = (1, 1)$ is an equilibrium, i.e. $Z_1(1, 1) = Z_2(1, 1) = 0$.
2. Compute the Jacobian $J_Z(p_1, p_2)$ and evaluate it at $(1, 1)$.
3. Interpret each entry of $J_Z(1, 1)$ economically. In particular: what does $\partial Z_1 / \partial p_2 > 0$ tell you about how the two goods are related?
4. Compute the directional derivative of Z_1 at $(1, 1)$ in the direction $v = (1, 1) / \sqrt{2}$ (a simultaneous equal rise in both prices). Interpret the result: what happens to excess demand for good 1 when all prices scale up equally?
5. Verify that $p_1 Z_1(p_1, p_2) + p_2 Z_2(p_1, p_2) = 0$ for all (p_1, p_2) . This is known as **Walras' Law**: the total value of excess demand across all goods is always zero. Give an economic interpretation: why does it make sense that consumers, taken together, cannot be simultaneously trying to buy more than they can afford?

Solution

Question 1.

$$Z_1(1, 1) = \frac{1}{1} - 1 = 0, \quad Z_2(1, 1) = \frac{1}{1} - 1 = 0. \quad \checkmark$$

Both markets clear at equal prices.

Question 2.

Compute all four partial derivatives:

$$\frac{\partial Z_1}{\partial p_1} = -\frac{p_2}{p_1^2}, \quad \frac{\partial Z_1}{\partial p_2} = \frac{1}{p_1}, \quad \frac{\partial Z_2}{\partial p_1} = \frac{1}{p_2}, \quad \frac{\partial Z_2}{\partial p_2} = -\frac{p_1}{p_2^2}.$$

The Jacobian at a general point is:

$$J_Z(p_1, p_2) = \begin{pmatrix} -p_2/p_1^2 & 1/p_1 \\ 1/p_2 & -p_1/p_2^2 \end{pmatrix}.$$

Evaluating at $(1, 1)$:

$$J_Z(1, 1) = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Question 3.

Each entry has a direct economic interpretation:

- $\partial Z_1/\partial p_1 = -1 < 0$: when good 1 becomes more expensive, excess demand for it falls. This is the **law of demand**.
- $\partial Z_1/\partial p_2 = 1 > 0$: when good 2 becomes more expensive, consumers substitute toward good 1, raising excess demand for it. The two goods are **substitutes**.
- $\partial Z_2/\partial p_1 = 1 > 0$: symmetrically, good 2 is also a substitute for good 1.
- $\partial Z_2/\partial p_2 = -1 < 0$: the law of demand for good 2.

Note: symmetry is not always satisfied, look at the more general jacobian to see it.

Question 4.

The directional derivative of Z_1 at $(1, 1)$ in direction $v = (1, 1)/\sqrt{2}$ is:

$$D_v Z_1(1, 1) = \nabla Z_1(1, 1) \cdot v = (-1, 1) \cdot \frac{1}{\sqrt{2}}(1, 1) = \frac{1}{\sqrt{2}}(-1 + 1) = 0.$$

A proportional rise in all prices leaves excess demand unchanged. This reflects a fundamental property: only **relative prices** determine demand, not the overall price level. Doubling all prices does not change what anyone wants to buy or sell. This is known as **homogeneity of degree zero**.

Question 5.

Verification:

$$p_1 Z_1 + p_2 Z_2 = p_1 \left(\frac{p_2}{p_1} - 1 \right) + p_2 \left(\frac{p_1}{p_2} - 1 \right) = p_2 - p_1 + p_1 - p_2 = 0. \quad \checkmark$$

Economic interpretation. Think of the consumers collectively as having a fixed budget. When they decide how much of each good to buy, their total spending on all goods equals their total income: they don't throw money away (which property of utility am I implicitly relying on?), and they can't spend more than they have. Walras' Law is the aggregate version of this budget constraint: the value of the goods consumers want to buy in excess ($Z_i > 0$) is exactly offset by the value of the goods in which there is a surplus ($Z_i < 0$). The economy as a whole is always "on budget." A useful consequence: if $n - 1$ markets are in equilibrium, the n -th market must be too.

Exercise 11 : Chain rule and the Solow growth decomposition (difficult)

In macroeconomics, a **production function** maps inputs to output. The **Cobb-Douglas** production function is widely used:

$$Y(K, L) = K^\alpha L^{1-\alpha}, \quad \alpha \in (0, 1),$$

where Y is output (GDP), K is capital, L is labor, and α is capital's share of income.

Both K and L grow over time:

$$K(t) = K_0 e^{g_K t}, \quad L(t) = L_0 e^{g_L t},$$

where $g_K > 0$ and $g_L > 0$ are the growth rates of capital and labor respectively.

1. Use the **chain rule** to compute $\frac{dY}{dt}$ as a function of $K(t)$, $L(t)$, g_K , g_L , and α .

2. Define the **growth rate of output** as $g_Y = \frac{\dot{Y}}{Y} = \frac{1}{Y} \frac{dY}{dt}$. Show that:

$$g_Y = \alpha g_K + (1 - \alpha) g_L.$$

3. This result is the **Solow growth decomposition** (after Nobel laureate Robert Solow, who developed it in the 1950s). Interpret it as follows: in a competitive economy, firms pay each input its marginal product, so the wage bill is $(1 - \alpha)Y$ and the capital income is αY . Explain why α and $(1 - \alpha)$ appear as weights, and what the decomposition says about how growth in each input contributes to output growth.

4. Suppose capital grows at $g_K = 3\%$ and labor at $g_L = 1\%$, with $\alpha = 1/3$. Compute the implied output growth rate g_Y . Break down the contribution of each factor explicitly.

5. In practice, observed output growth is often **higher** than $\alpha g_K + (1 - \alpha)g_L$. Solow called this gap **Total Factor Productivity (TFP)** growth, or the **Solow residual**. If actual output growth is $g_Y = 2.5\%$ using the numbers from Question 4, what is the TFP growth rate? What does TFP capture that capital and labor growth do not?

i Solution

Question 1.

Apply the chain rule, treating K and L as functions of t :

$$\frac{dY}{dt} = \frac{\partial Y}{\partial K} \cdot \frac{dK}{dt} + \frac{\partial Y}{\partial L} \cdot \frac{dL}{dt}.$$

Compute each piece:

$$\begin{aligned} \frac{\partial Y}{\partial K} &= \alpha K^{\alpha-1} L^{1-\alpha}, & \frac{dK}{dt} &= g_K K. \\ \frac{\partial Y}{\partial L} &= (1 - \alpha) K^\alpha L^{-\alpha}, & \frac{dL}{dt} &= g_L L. \end{aligned}$$

Substituting:

$$\frac{dY}{dt} = \alpha K^{\alpha-1} L^{1-\alpha} \cdot g_K K + (1 - \alpha) K^\alpha L^{-\alpha} \cdot g_L L = \alpha g_K K^\alpha L^{1-\alpha} + (1 - \alpha) g_L K^\alpha L^{1-\alpha}.$$

Factoring out $Y = K^\alpha L^{1-\alpha}$:

$$\frac{dY}{dt} = [\alpha g_K + (1 - \alpha) g_L] Y.$$

Question 2.

Dividing both sides by Y :

$$g_Y = \frac{\dot{Y}}{Y} = \alpha g_K + (1 - \alpha) g_L. \quad \checkmark$$

Question 3.

The formula says output growth is a **weighted average** of capital growth and labor growth, with weights α and $1 - \alpha$. These weights have a precise economic meaning: under competitive markets, the marginal product of capital is $\alpha Y/K$ and firms hire capital until

this equals the rental rate. Aggregating over the whole economy, capital earns a share α of total output and labor earns a share $1 - \alpha$. So the weight attached to each input's growth rate is exactly its share of income — the more economically important an input, the more its growth rate matters for output.

A 1% increase in capital raises output by $\alpha\%$, and a 1% increase in labor raises output by $(1 - \alpha)\%$. Solow's insight (1956) was that this formula, combined with data on capital and labor growth, leaves a **residual** — the part of output growth not explained by factor accumulation — which he attributed to **technological progress**.

Question 4.

$$g_Y = \frac{1}{3} \times 3\% + \frac{2}{3} \times 1\% = 1\% + \frac{2}{3}\% \approx 1.67\%.$$

Capital contributes 1% and labor contributes 0.67%. The relatively small weight on capital ($\alpha = 1/3$) means that even though capital is growing faster, labor growth — because workers represent two thirds of the economy — accounts for more of total output growth in this example.

Question 5.

The Solow residual (TFP growth) is the gap between actual output growth and the growth accounted for by inputs:

$$\text{TFP growth} = g_Y - \alpha g_K - (1 - \alpha)g_L = 2.5\% - 1.67\% = 0.83\%.$$

TFP captures everything that makes the economy more productive *beyond* simply having more capital and labor: technological innovation, improvements in the quality of management, better allocation of resources across firms, and so on. Because it is computed as a residual rather than measured directly, Solow himself called it “a measure of our ignorance.” In modern growth accounting, separating TFP from factor accumulation is one of the central empirical challenges.

Exercise 12 : Implicit function theorem: money market equilibrium

Implicit Function Theorem (scalar case)

Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ be C^1 , and suppose $F(x_0, y_0) = 0$ with $\frac{\partial F}{\partial y}(x_0, y_0) \neq 0$. Then near (x_0, y_0) there exists a unique C^1 function $y = g(x)$ such that $F(x, g(x)) = 0$, and:

$$g'(x_0) = - \frac{\partial F / \partial x}{\partial F / \partial y} \Big|_{(x_0, y_0)}.$$

The **money market** is in equilibrium when money demand equals money supply. A standard specification is:

$$M = P \cdot L(Y, i),$$

where:

- $M > 0$ is the (fixed) **money supply**, controlled by the central bank
- $P > 0$ is the **price level**
- $Y > 0$ is **real output** (GDP)
- $i > 0$ is the **nominal interest rate**
- $L(Y, i) = \frac{Y}{i}$ is **real money demand**: higher output raises demand, higher interest rates reduce it (since bonds become more attractive)

Rewrite the equilibrium condition as:

$$F(Y, i; M, P) \equiv \frac{PY}{i} - M = 0.$$

A note on realism. The functional form $L(Y, i) = Y/i$ is a deliberate simplification. Empirical money demand functions typically take the form $L(Y, i) = Y^\beta/i^\gamma$ with $\beta, \gamma > 0$, and the log-linearized version $\ln L = \beta \ln Y - \gamma \ln i$ is standard in monetary economics. The unit elasticities ($\beta = \gamma = 1$) assumed here make the algebra clean while preserving the correct qualitative properties: money demand rises with output and falls with the interest rate. The IFT results below are qualitatively robust to more general specifications. Moreover, for the sake of simplicity, we will work with $i = 1$ (a 100% interest rate)...

1. Verify that the point $(Y_0, i_0) = (M/P, 1)$ satisfies $F = 0$.
2. Check that the IFT condition $\frac{\partial F}{\partial i} \neq 0$ holds at (Y_0, i_0) . Why is this the relevant partial derivative? (*Think about which variable we want to express as a function of which.*)
3. Apply the IFT to compute $\frac{\partial i^*}{\partial Y}$: how does the equilibrium interest rate respond to a rise in output? This relationship traces out the **LM curve**.
4. Apply the IFT to compute $\frac{\partial i^*}{\partial M}$: how does the interest rate respond to an increase in the money supply? Provide economic intuition.
5. (*Harder.*) Now treat both Y and M as varying simultaneously. Use the IFT formula

$$\begin{pmatrix} \partial i^* / \partial Y \\ \partial i^* / \partial M \end{pmatrix} = - \left(\frac{\partial F}{\partial i} \right)^{-1} \begin{pmatrix} \partial F / \partial Y \\ \partial F / \partial M \end{pmatrix}$$

to recover your answers from parts 3 and 4 in a single computation. (*Note: since F maps to $\mathbb{R}$, the “matrix” $\partial F / \partial i$ is just a scalar, and its inverse is simply its reciprocal. This is the scalar special case of a more general matrix IFT you will encounter later in the course.*)

i Solution

Question 1.

Substitute $(Y_0, i_0) = (M/P, 1)$ directly:

$$F\left(\frac{M}{P}, 1\right) = \frac{P \cdot (M/P)}{1} - M = M - M = 0. \quad \checkmark$$

Question 2.

Compute $\frac{\partial F}{\partial i}$:

$$\frac{\partial F}{\partial i} = -\frac{PY}{i^2}.$$

At $(Y_0, i_0) = (M/P, 1)$:

$$\left. \frac{\partial F}{\partial i} \right|_{(M/P, 1)} = -\frac{P \cdot (M/P)}{1^2} = -M \neq 0 \quad (\text{since } M > 0).$$

We check $\partial F / \partial i$ because we want to express i (the interest rate) as a function of Y (output). The IFT requires the partial of F with respect to the **dependent variable** i to be nonzero at the reference point.

Question 3.

Compute $\frac{\partial F}{\partial Y} = \frac{P}{i}$. By the IFT:

$$\frac{\partial i^*}{\partial Y} = -\frac{\partial F/\partial Y}{\partial F/\partial i} = -\frac{P/i}{-PY/i^2} = \frac{i}{Y} > 0.$$

Interpretation. A rise in output Y increases money demand $L(Y, i) = Y/i$. With fixed money supply M , the market can only clear if the interest rate rises (making money demand fall back). This is the standard explanation for the **upward slope of the LM curve**.

Question 4.

Compute $\frac{\partial F}{\partial M} = -1$. By the IFT:

$$\frac{\partial i^*}{\partial M} = -\frac{\partial F/\partial M}{\partial F/\partial i} = -\frac{-1}{-PY/i^2} = -\frac{i^2}{PY} < 0.$$

Interpretation. An increase in the money supply M creates excess supply of money at the prevailing interest rate. Holders of excess money buy bonds, driving bond prices up and interest rates down. This is the standard **expansionary monetary policy** result: more money in the economy lowers the cost of borrowing.

Question 5.

Recall $\frac{\partial F}{\partial i} = -PY/i^2$, so $(\frac{\partial F}{\partial i})^{-1} = -\frac{i^2}{PY}$. Substituting:

$$\begin{pmatrix} \partial i^*/\partial Y \\ \partial i^*/\partial M \end{pmatrix} = - \begin{pmatrix} i^2 \\ -PY \end{pmatrix} \begin{pmatrix} P/i \\ -1 \end{pmatrix} = \frac{i^2}{PY} \begin{pmatrix} P/i \\ -1 \end{pmatrix} = \begin{pmatrix} i/Y \\ -i^2/(PY) \end{pmatrix}.$$

This recovers exactly the results from Questions 3 and 4 in a single computation. ✓